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# Acetaminophen Toxicity

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## Continuing Education Activity

Acetaminophen is one of the most commonly used analgesics and antipyretics. Although generally safe at therapeutic doses, acetaminophen poisoning causes hepatic necrosis in overdose. Acetaminophen toxicity is the second most common reason for liver transplantation worldwide and the most common cause of acute liver failure in the United States. Responsible for 56,000 emergency department visits and 2600 hospitalizations, acetaminophen poisoning causes 500 deaths annually in the United States. Approximately 50% of these poisonings are unintentional, often resulting from patients misinterpreting dosing instructions or unknowingly consuming multiple acetaminophen-containing products.

The mortality associated with acetaminophen overdose is low if recognized and treated within the first 8 hours after an acute ingestion. Factors such as alcohol use, genetics, age, medications, use of herbal supplements, and nutritional status may affect the prognosis. While hepatic injury is the most common effect of acetaminophen poisoning, acute renal injury is also common. This activity reviews the etiology, evaluation, treatment, and management of acetaminophen poisoning while highlighting the use of interprofessional teams to manage and prevent this problem.

**Objectives:**

- Implement effective screening protocols to identify patients at risk of acetaminophen poisoning in various healthcare settings.
- Differentiate between acetaminophen toxicity and other potential causes of liver dysfunction, ensuring accurate and targeted interventions.
- Select evidence-based strategies to ensure timely initiation of treatment for acetaminophen poisoning and promote optimal patient outcomes.
- Collaborate with hepatologists, critical care specialists, toxicologists, psychiatrists, and pharmacists to optimize the management of acetaminophen poisoning and enhance overall patient care.

Access free multiple choice questions on this topic.

## Introduction

Acetaminophen is responsible for more pharmaceutical overdoses than any other medication in the United States and worldwide, primarily due to the drug's widespread availability and the common perception of its safety. In the United States, acetaminophen toxicity is the leading cause of acute liver failure. Over 60 million Americans use acetaminophen weekly, often unknowingly, due to its presence in various combination products, particularly those containing opioids and diphenhydramine.[1][2]**[3]** Acetaminophen, also known as *N*-acetyl-para-aminophenol or paracetamol, functions as an antipyretic and analgesic with a mechanism distinct from typical nonsteroidal anti-inflammatory medications. Although its mode of action, specifically the selective inhibition of cyclooxygenase in the brain, is not fully elucidated, acetaminophen effectively alleviates fever and pain. Acetaminophen may also inhibit

prostaglandin synthesis in the central nervous system and directly affect the hypothalamus to produce an antipyretic effect.[4]

If left untreated, acetaminophen toxicity can lead to fatal hepatic necrosis. Timely intervention is essential in preventing fulminant liver failure and the need for liver transplantation. Activated charcoal can be considered in the first 2 hours after an acute ingestion, although there are limited data supporting improved outcomes. Acetylcysteine, most often given IV, is the critical antidote for acetaminophen poisoning, and is most effective if administered within the first 8 hours after an acute overdose. Patients with suspected chronic acetaminophen poisoning should receive acetylcysteine as soon as possible. The most reliable predictor of acute toxicity is correlating the time of ingestion with the serum acetaminophen concentration via the Rumack-Matthew nomogram. Patients whose serum acetaminophen concentrations are above the treatment line require acetylcysteine. A striking 50% of cases result from unintentional overdoses, highlighting a crucial need for healthcare professionals to properly educate patients regarding the proper dosing of acetaminophen and its presence in prescribed and over-the-counter preparations.

## Etiology

Acetaminophen is available in immediate-release and extended-release formulations. The dosage forms available are capsules, elixir, gel, oral and intravenous solutions, liquid, tablets, suppositories, syrup, and dissolve packs. When taken at therapeutic doses, acetaminophen has a good safety profile. The therapeutic doses are:

- 10 to 15 mg/kg/dose in children every 4 to 6 hours with a maximum dose of 75 mg/kg/d
- 325 to 1000 mg/dose in adults every 4 to 6 hours, with a maximum daily dose of 3 g/d

Toxicity is likely to develop in adults at:

- Doses greater than 12 g over 24 hours
- Single doses of 7.5 to 10 g
- Single doses greater than 150 mg/kg

Toxicity in children occurs following a single dose of 150 mg/kg or 200 mg/kg in otherwise healthy children aged 1 to 6.

Typical doses of acetaminophen are fully absorbed within 2 hours of ingestion. Foods and medications that decrease gastric emptying can delay absorption. The liver metabolizes 89% of acetaminophen, predominantly via phase II glucuronidation and sulfation. At higher doses and as these pathways saturate, CYP oxidation of acetaminophen forms a toxic, oxidative metabolite and depletes hepatic glutathione stores.

## Epidemiology

Acetaminophen toxicity accounts for 50% of all reported cases of liver failure in the United States and 20% of all liver transplants. If identified and treated rapidly, the associated morbidity and mortality are low. Once liver failure develops, the mortality increases to 28%, with one-third requiring a liver transplant. Although acetaminophen poisoning is more common in children, adults are more likely to develop hepatotoxicity. Acetaminophen poisoning is responsible for 56,000 emergency department visits, 2600 hospitalizations, and 500 deaths annually in the United States, with 50% of these cases being unintentional overdoses.[5][6][7] Approximately 30,000 pediatric acetaminophen poisoning cases are reported to the National Poison Data System annually.

Factors that increase the risk of hepatotoxicity include underlying liver disease, alcohol use disorder, and unintentional overdose. Combination opioid prescriptions containing

acetaminophen are an additional risk contributing to acetaminophen poisoning.

## Pathophysiology

Acetaminophen is rapidly absorbed from the gastrointestinal tract and reaches therapeutic levels in 30 minutes to 2 hours. In overdose, serum concentrations peak at approximately 4 hours unless other factors are included, such as the co-ingestion of an agent that slows gastric motility or if the acetaminophen is in an extended-release form.[8][9] Acetaminophen has an elimination half-life of 4 hours.

**Metabolism**

Approximately 90% of acetaminophen is metabolized in the liver by the hepatic microsomes via sulfotransferases (SULTs) and UDP-glucuronosyl transferases, forming nontoxic sulfate and glucuronide conjugates that are excreted in the urine.[10][11]

At therapeutic doses, approximately 8% of acetaminophen is metabolized by hepatic cytochrome P450 subfamilies—particularly CYP2E1—through the mixed-function oxidase pathway, generating a toxic, highly reactive, electrophilic intermediate *N*-acetyl-p-benzoquinoneimine (NAPQI).

The remaining 2% of acetaminophen is excreted unchanged in the urine.

At therapeutic doses, acetaminophen produces minimal amounts of NAPQI, which swiftly conjugates with hepatic glutathione to produce nontoxic cysteine and mercaptate compounds excreted in the urine. However, in cases of acetaminophen toxicity, an increased production of NAPQI occurs, depleting hepatic glutathione stores.

NAPQI gains an additional aryl group and binds covalently to cysteine groups on hepatic macromolecules, particularly mitochondrial proteins, forming NAPQI-protein adducts. The process initiates an irreversible cascade. The results of NAPQI-protein adduct formation include:

- Oxidative hepatocyte injury
- Alteration of the mitochondrial ATP-synthase alpha-subunit
- Hepatocellular centrilobular (zone 3) necrosis

Processes that contribute to the propagation of hepatic injury include:

- Generation of nitrotyrosine adducts within mitochondria by toxic free radicals, such as peroxynitrite
    - Results in damage to mitochondrial DNA and ATP-synthase, halting ATP synthesis
- Lipid peroxidation and membrane damage
- Release of cytokines, apoptosis-inducing factor, endonuclease G, and reactive nitrogen and oxygen species from damaged mitochondria
    - Causes a secondary inflammatory response
- Recruitment of inflammatory cells through the innate immune system by damage-associated molecular pattern (DAMP) products, such as nuclear fragments and mitochondrial DNA
    - Typically manifests during clinical stage II

**Clinical Factors Affecting Toxicity**

Several factors can alter the risk of acetaminophen-induced liver damage. In some cases, patients may develop subclinical toxicity following repeated therapeutic dosing.

- Acute alcohol ingestion: The risk of hepatotoxicity may *decrease* when acetaminophen is consumed concurrently with alcohol. Both acetaminophen and alcohol serve as substrates for the CYP2E1 enzyme. A study suggests a 10% reduction in hepatic toxicity, possibly attributed to reduced NAPQI production.[12][13]
- Chronic alcohol ingestion: Despite upregulation of CYP2E1 and depletion of glutathione stores and synthesis, there is minimal evidence to indicate that chronic alcohol ingestion causes an increased incidence of hepatic toxicity with therapeutic doses of acetaminophen.[12] Chronic alcohol use may, however, increase the risk of hepatotoxicity with repeated supratherapeutic acetaminophen doses. The upregulation of CYP2E1, combined with the increased risk of malnutrition, recent fasting, and low glutathione stores, predisposes patients with alcohol use disorder to hepatotoxicity from repeated supratherapeutic doses.
- Chronic liver disease: Patients with chronic liver disease who abstain from alcohol do not face an elevated risk of hepatotoxicity when ingesting acetaminophen. Individuals with cirrhosis may exhibit lower CYP450 enzyme activity, including CYP2E1, which may offer some protection in the context of acetaminophen overdose.
- Concurrent medications and herbal supplements: Medications that induce CYP2E1 enzymes, such as carbamazepine or phenobarbital, may increase the risk of hepatotoxicity. Medications that compete for glucuronidation pathways, such as trimethoprim-sulfamethoxazole and opioids, may also result in a higher risk. Herbal supplements such as St John's wort, garlic, and germander may enhance the CYP450 activity.
- Nutritional status: Hepatic glucuronidation depends on liver carbohydrate reserves. Fasting or malnutrition decreases the glucuronidation of acetaminophen, causing an increased production of NAPQI.
- Age: The risk of hepatotoxicity increases with age. Children younger than 5 appear less susceptible to hepatotoxicity. The increased supply and regeneration of glutathione and greater sulfation capacity in children are likely reasons for the lower risk of hepatotoxicity.
- Genetics: The link to genetic polymorphisms that cause acetaminophen's diminished or excessive oxidative metabolism is unclear.
- Tobacco use: Tobacco use induces CYP1A2 enzymes and is an independent risk factor for mortality after acetaminophen overdose. The highest mortality is associated with patients who smoke and drink alcohol.[13]

## Histopathology

The histological features of acetaminophen toxicity reveal cytolysis and centrilobular necrosis. These changes result from an increased concentration of CYP2E1 enzymes near the central vein, causing a high local concentration of NAPQI.[14] Clinical recovery precedes histopathological recovery. Histopathologic recovery may take up to 3 months, but when it occurs, it is complete. Residual hepatic insufficiency is not a long-term sequelae of acetaminophen poisoning.

## History and Physical

Most patients with acetaminophen toxicity are initially either asymptomatic or present with nausea and vomiting. Obtaining a detailed history, including accurate timing, formulation, dosage, and any coingestants, is essential for proper diagnosis and management. The clinical course of acetaminophen toxicity classically consists of 4 stages.[15]

**Stage I**

- First 30 minutes to 24 hours

- Possibly asymptomatic
- May experience nausea, vomiting, diaphoresis, pallor, lethargy, and malaise
- Transaminase values are generally normal, but may increase within 8 to 12 hours after larger ingestions
- Central nervous system depression and elevated anion gap metabolic acidosis are possible with larger ingestions

**Stage II**

- 24 to 72 hours
- Onset of liver injury, manifesting as transaminase elevation
- Clinically, the patient may appear improved, but laboratory values worsen
- Right upper quadrant pain and hepatomegaly are possible
- Synthetic liver function markers (eg, prothrombin time, bilirubin) may begin to rise
- Possible renal injury and/or oliguria
- Case reports of acute pancreatitis

**Stage III**

- 72 to 96 hours
- Aminotransferase levels peak, with values often exceeding 10,000 IU/L
- More pronounced symptoms, including jaundice, encephalopathy, hyperammonemia, abnormal aminotransferases, and bleeding diathesis, along with possible recurrence of symptoms from stage 1
- Acute renal failure is common, particularly in patients with liver failure
- Indirect hyperbilirubinemia
- Prolonged PT
- Hypoglycemia
- Lactic acidosis
- The most common stage for death to occur due to multisystem organ failure

**Stage IV**

- If the patient survives stage 3
- Recovery phase
- Typically begins by day 4 and completes by day 7
- Symptoms and laboratory values may require several weeks to normalize
- Histopathologic changes in the liver become evident
- No long-term liver dysfunction or cirrhosis

Acute renal failure due to acute tubular necrosis manifests as an elevated blood urea nitrogen and creatinine, proteinuria, hematuria, and granular and epithelial cell casts on urinalysis. The incidence is less than 2% of all patients, 5% of cases with liver involvement but no hepatic

failure, 10% of severe poisonings, and nearly 53% of cases with acute hepatic failure. The renal function generally returns to normal in 1 to 4 weeks. Some patients may require dialysis during this time.

## Evaluation

**Overall Approach**

Relating the time of initial ingestion to the serum acetaminophen concentration provides a more accurate prediction of the risk of hepatotoxicity than relying on the dose history.

The initial evaluation begins with the following:

- Identifying the agents involved
- Assessing severity
- Predicting the toxicity

Essential historical information to gather includes:

- Elicit the dose
- Elicit the intent
- Pattern of use
- Time of ingestion
- Presence of co-ingestants
- Presence of comorbid conditions such as Gilbert disease, alcohol use disorder, or epilepsy with the use of anti-seizure medications

The diagnosis of acetaminophen toxicity begins with obtaining a serum acetaminophen concentration, regardless of symptoms. If the timing of ingestion is unclear, an acetaminophen concentration should be obtained at the time of presentation. Additional recommended investigations include:

- Serum aminotransferases
- PT and international normalized ratio (INR)
- Electrolytes
- Blood urea nitrogen and creatinine
- Serum total bilirubin concentration
- Human chorionic gonadotropin in women of childbearing age
- Arterial or venous blood gas and serum lactate in critically ill patients or those with altered mental status
- Electrocardiogram in patients with an intentional overdose
- Serum salicylate concentration

**Evaluation After Ingestion of Immediate-Release Acetaminophen (most formulations)**

- Obtain a serum acetaminophen concentration 4 hours after the ingestion
- Obtain serum acetaminophen concentration immediately if the ingestion was more than 4 hours before presentation.

- Determine the need for *N*-acetylcysteine based on the Revised Rumack-Matthew nomogram.[16][17][18]
- Administer empiric *N*-acetylcysteine if ingestion occurred more than 8 hours before presentation

If the timing of ingestion is unknown, *N*-acetylcysteine should be started immediately while awaiting the following laboratory tests:

- Serum acetaminophen concentration
- Electrolytes
- Blood urea nitrogen and creatinine
- Serum total bilirubin
- Serum transaminases

A nondetectable serum acetaminophen concentration drawn between 2 and 4 hours after ingestion excludes toxicity.

**Evaluation After Ingestion of Extended-Release Formulation or Co-Ingestion with Anticholinergics or Opioids**

An extended-release formulation of acetaminophen is available in some regions. Clinicians use the Revised Rumack-Matthew nomogram to assess patients who ingest an extended-release formulation or coingest anticholinergic or opioid medications, which may slow absorption of acetaminophen. If the initial 4-hour acetaminophen concentrationis ≤10 mcg/mL, a repeat level is unnecessary. If the acetaminophen level between 4 and 12 hours is below a level indicating the need for acetylcysteine but >10 mcg/mL (66 µmol/L), another acetaminophen level should be obtained in 4 to 6 hours.

**Evaluation After Repeated Supratherapeutic Ingestion**

Repeated supratherapeutic ingestions are multiple ingestions that occur over more than 24 hours. The Rumack-Matthew nomogram cannot be used in this scenario. The decision to initiate *N*-acetylcysteine therapy should be based on a thorough history, physical examination, and laboratory evaluation. If a supratherapeutic ingestion of acetaminophen is suspected, *N*-acetylcysteine should be started immediately while obtaining laboratory studies. If the serum acetaminophen concentration is undetectable and aminotransferases are within normal limits, acetaminophen toxicity can be excluded, and *N*-acetylcysteine can be discontinued.

## Treatment / Management

The choice to use activated charcoal for acetaminophen poisoning is controversial. Some studies suggest using activated charcoal if the patient presents within 4 hours of ingestion, as it has been shown to decrease acetaminophen absorption, reduce the need for *N*-acetylcysteine, and lower the risk of liver injury when administered within 4 hours of ingestion.[19][20][21][22] Use beyond 1 hour is most effective with large ingestions or with ingestion of an extended-release form or any medication that can delay gastric emptying. Contraindications to activated charcoal include gastrointestinal obstruction and an unprotected airway. Orogastric lavage or whole bowel irrigation is not recommended.[23][24][25] Further, as *N*-acetylcysteine is known to be very effective in the prevention of hepatotoxicity if administered early, activated charcoal may not be necessary in most ingestions.

**Four to Eight Hours After Ingestion**

The serum acetaminophen level should be plotted on the Revised Rumack-Matthew nomogram. If the patient's serum acetaminophen concentration falls at or above the treatment line, *N*-acetylcysteine administration is indicated. The treatment threshold is 150 mcg/mL or 990 µmol/L at 4 hours and 4.69 mcg/mL or 31.3 µmol/L at 24 hours. Serum concentrations above this threshold place the patient at risk for hepatotoxicity, and treatment with *N*-acetylcysteine is indicated. Patients who have ingested an extended-release formulation or co-ingested an opiate or anticholinergic medication may need a repeat acetaminophen level in 4 to 6 hours.

**More than Eight Hours After Ingestion**

In cases of suspected acute ingestion exceeding 150 mg/kg or a total dose of 7.5 g, and when serum acetaminophen levels are unavailable until after 8 hours from ingestion, healthcare professionals should initiate *N*-acetylcysteine therapy while awaiting acetaminophen concentrations. Beyond 8 hours, the risk of liver injury increases as the time to administration of *N*-acetylcysteine increases. Discontinuation of *N*-acetylcysteine is appropriate when acetaminophen is no longer detectable, and liver function tests are normal.

Patients with repeated supratherapeutic ingestions with a detectable serum acetaminophen concentration *or* elevated aminotransferases warrant *N*-acetylcysteine. *N*-acetylcysteine should be administered to patients with any evidence of liver injury and a history of acetaminophen ingestion. Patients can still benefit from *N*-acetylcysteine after the initial ingestion or until acetaminophen is no longer detectable in the serum while aminotransferases remain elevated. *N*-acetylcysteine is an antioxidant that diminishes hepatic necrosis, decreases neutrophil infiltration, improves microcirculatory blood flow, and increases tissue oxygen delivery. Intermittent hemodialysis effectively removes acetaminophen, but is only indicated in unique cases of very large ingestions.

Acetaminophen, AST, ALT, and INR should be measured every 12 hours following the initiation of *N*-acetylcysteine.[26] If the AST or ALT exceeds 1000 IU/L, encephalopathy should be monitored, and serum bicarbonate, glucose, and creatinine should be measured every 12 hours.

Dialysis may be necessary if the patient develops acute kidney injury. Indications for dialysis include refractory fluid overload, severe hyperkalemia, severe metabolic acidosis, and uremia. Dose adjustments for patients with alcohol use disorder or those who are chronically ill are not required. Dialysis is considered safe for use during pregnancy.

**High-Risk Ingestion**

Ingestions of greater than 30 g or with a 4-hour concentration of greater than 300 mcg/mL are at higher risk of liver injury. These patients may require intubation to protect their airway, intravenous fluids and vasopressors, and bicarbonate for metabolic acidosis. *N*-acetylcysteine should be initiated immediately. In addition to *N*-acetylcysteine, hemodialysis is indicated in patients with an acetaminophen concentration >900 mcg/mL.

***N*-acetylcysteine Indications**

*N*-acetylcysteine is extremely effective in the prevention of liver injury if administered within 8 hours of an acute acetaminophen ingestion. The effectiveness remains unchanged whether initiated between 0 and 4 hours or 4 and 8 hours. *N*-acetylcysteine acts by preventing the binding of NAPQI to hepatic macromolecules, acting as a substitute for glutathione, serving as a precursor for endogenous antioxidants, and reducing oxidative NAPQI. Indications for *N*-acetylcysteine include:

- A serum acetaminophen concentration above the toxic threshold according to the Revised Rumack-Matthew nomogram
- An acetaminophen concentration greater than 10 mcg/mL with an unknown time of ingestion

- A dose of acetaminophen >140 mg/kg ingested more than 8 hours before presentation
- Ingestion with any evidence of liver injury [27]

Either oral or intravenous (IV) *N*-acetylcysteine is acceptable. Intravenous *N*-acetylcysteine is preferred for patients with intractable vomiting, those who refuse oral intake, those at risk for aspiration, or those who are pregnant or have fulminant liver failure. The IV formulation may decrease the length of the hospital stay and may be tolerated better by the patient, as the oral form has an unfavorable taste and odor. The oral formulation also requires 18 doses given 4 hours apart, with the total treatment time being 72 hours. In comparison, the IV formulation may require as little as 21 hours of treatment.

**Oral *N*-acetylcysteine**: The oral regimen consists of 18 doses administered 4 hours apart, resulting in a total treatment time of 72 hours. The protocol begins with administering a 140 mg/kg oral loading dose followed by 70 mg/kg every 4 hours until achieving the stop criteria. Patients weighing over 100 kg receive a maximum 100 mg/kg dose.

**Intravenous *N*-acetylcysteine**: Approximately 5% of patients may experience non–IgE-mediated anaphylactoid reactions to *N*-acetylcysteine with the IV formulation, particularly during the bolus.

Several IV protocols have been studied and implemented:

- 21-hour intravenous "3-bag regimen":
- An initial loading dose of 150 mg/kg IV is administered over 60 minutes.
- Next, 50 mg/kg is administered over 4 hours, or 12.5 mg/kg/hr IV for 4 hours.
- Finally, a dose of 100 mg/kg is administered over 16 hours or 6.25 mg/kg/h IV for 16 hours.
- Serum acetaminophen concentrations and transaminases should be obtained 2 hours prior to completion of the third dose. If the acetaminophen concentration is detectable *or* transaminases are elevated, *N*-acetylcysteine should be continued until acetaminophen is undetectable and transaminases are normal or have significantly declined from peak.
- Simplified "one-bag" regimen[28][29][28]:
- An initial loading dose of 150 mg/kg IV is administered over 60 minutes.
- A continuous infusion of 12.5 - 15 mg/kg/hr is administered until the serum acetaminophen concentration is undetectable and transaminases are normal or have significantly declined from peak.

Other protocols have been studied and used worldwide. In cases of high-risk ingestions, where reports indicate liver injury with ingestions exceeding 50 g or serum acetaminophen concentrations surpassing 500 mg/L or 3300 µmol/L, some clinicians suggest administering a higher dose of *N*-acetylcysteine. This only applies to the use of a "3-bag regimen" or any regimen where the final dose is 6.25 mg/kg/hr. A variety of approaches have been adopted. [30] Consulting a poison center or medical toxicologist is advisable. If not available, a reasonable alternative is to increase the final infusion rate of the 21-hour protocol to 12.5 mg/kg/h. Patients undergoing hemodialysis do not need an adjustment of oral *N*-acetylcysteine, but IV *N*-acetylcysteine infusion rates are 12.5 mg/kg/h during hemodialysis.

**Stopping Criteria**

The definitive duration of *N*-acetylcysteine therapy is a matter of controversy. Although no universally accepted stopping criteria exist, the decision to discontinue *N*-acetylcysteine should be based on laboratory evidence of improvement, rather than solely on the duration of therapy.

[26] A recent consensus statement recommends continuing *N*-acetylcysteine at a rate of at least 6.25 mg/kg/h or 70 mg/kg every 4 hours orally until the serum acetaminophen level is less than 10 mcg/mL, INR <2.0, the ALT and AST are normal for the patient or have decreased a minimum of 25% to 50% from their peak levels, and the patient is clinically well.[26] *N*-acetylcysteine may be beneficial beyond 72 hours if there is a fulminant hepatic failure until the patient receives a liver transplant or recovers.[31][32]

## Differential Diagnosis

The following list includes potential differential diagnoses:

- Hepatorenal syndrome
- Viral hepatitis
- Wilson disease
- Pancreatitis
- Acute tubular necrosis
- Amatoxin poisoning
- Cytomegalovirus infection
- Mononucleosis
- Gastroenteritis
- Peptic ulcer disease
- Other drug-induced or toxin-induced hepatitis
- Hepatobiliary disease
- Inborn errors of metabolism, including alpha1-antitrypsin deficiency or fatty acid oxidation abnormalities
- Reye syndrome
- Ischemic hepatitis

Clues to help differentiate acetaminophen as the cause of hepatitis include the acute onset and rapid progression with significant elevations of plasma aminotransferases (often >3000 IU/L) and its association with a rising PT and INR.

A significant increase in bilirubin levels is rare with acetaminophen toxicity. Unlike acetaminophen toxicity, transaminase elevations are typically mild in the case of alcohol-related hepatitis.

## Pertinent Studies and Ongoing Trials

Fomepizole is an antidote for toxic alcohol poisoning with methanol and ethylene glycol. This medication inhibits alcohol dehydrogenase and CYP2E1. Studies suggest that fomepizole may have a role as adjunct therapy along with *N*-acetylcysteine in patients determined to be at high risk of developing liver failure due to acetaminophen toxicity. Fomepizole prevents acetaminophen oxidation, inhibits Jun-N-terminal kinase, and decreases the formation of oxidative metabolites. Case reports of fomepizole are increasing, showing promise in critically ill patients.[33] Many guidelines do not currently recommend the routine use of fomepizole for acetaminophen toxicity, while others have begun implementing its use.[34][35][36] The role of fomepizole for acetaminophen poisoning is an ongoing area of study.

## Prognosis

The incidence of hepatotoxicity for patients treated with *N*-acetylcysteine within 8 hours of acetaminophen ingestion is very low, but it increases to approximately 40% when delayed beyond 16 hours. When treated promptly, the mortality associated with acetaminophen toxicity is less than 2%. However, if patients present late and develop severe liver failure, mortality is high. Approximately 1% to 3% of patients with severe liver failure require liver transplantation.[19][37][38] In general, children younger than 6 have a better prognosis than adults, primarily due to their greater capacity to detoxify acetaminophen via sulfation.

In patients with fulminant liver failure, the modified King's College Criteria helps to determine prognosis and provide insight as to which patients should receive care in a facility that offers liver transplants. King's College Criteria remains the preferred prognostic tool as compared to MELD and other clinical decision tools. The prognosis is poor without a liver transplant if the patient meets the following criteria:

- Arterial pH <7.3 after resuscitation, *or*:
- PT > 100, *and*
- Serum creatinine > 3.3, *and*
- Grade III or IV encephalopathy

Note that an arterial pH < 7.3 after resuscitation is, alone, considered an indication for consideration of liver transplant. The other 3 criteria must all be present to meet King's College Criteria. In clinical practice, patients approaching fulfillment of these criteria may benefit from early consultation with transplant teams, particularly if liver transplantation would require transfer to another healthcare center.

## Complications

Acetaminophen toxicity is associated with a variety of complications, ranging from mild to life-threatening, as follows:

- Acute tubular necrosis
- Hepatorenal syndrome
- Encephalopathy
- Life-threatening hemorrhage
- Liver failure
- Renal failure
- Death
- Stevens-Johnson syndrome
- Toxic epidermal necrolysis
- Acute generalized exanthematous pustulosis

## Postoperative and Rehabilitation Care

Patients may resume acetaminophen when they are clinically asymptomatic and have serum acetaminophen levels <10 mg/L, along with normal ALT, PT, and INR. Patients who survive without liver transplantation are not expected to have any long-term liver dysfunction or cirrhotic changes.

## Consultations

If a patient meets King's College Criteria or is nearing fulfillment of these criteria, consultation with a hepatologist and transplant surgeon is recommended. Consultation with a medical toxicologist or poison center is recommended for all patients with acetaminophen poisoning.

## Deterrence and Patient Education

Acetaminophen, also known as *N*-acetyl-para-aminophenol or paracetamol, is a widely used medication, both independently and in combination with numerous other medications worldwide. Despite its widespread use, acetaminophen is the leading cause of liver failure worldwide and the second most common indication for liver transplantation in the United States. With therapeutic use, acetaminophen is generally considered safe. However, instances of acetaminophen toxicity often arise due to patient misconceptions about dosing or a lack of awareness regarding its presence in multiple medications they may be consuming. Intentional ingestion of large doses also contributes to toxicity. Effective management of patients necessitates a thorough history of critical dosage details, formulations, and timing.

While decades of data have provided a strong understanding of acetaminophen toxicity and its treatment, patients achieve optimal outcomes when healthcare professionals intervene within the first 8 hours. Beyond this timeframe, mortality rates increase with each passing hour. The use of activated charcoal is controversial. Various factors, such as concurrent alcohol consumption, age, genetics, use of other medications, and use of herbal supplements, can affect the risk of acetaminophen toxicity. Complications extend beyond liver failure and may include kidney failure, encephalopathy, and death.

Treatment involves the administration of *N*-acetylcysteine, which helps prevent further liver injury. The duration of treatment typically spans 21 to 72 hours, depending on the specific formulation and protocol used. Some patients may require an extended treatment course based on the extent of liver damage. Clinicians monitor liver function tests and other laboratory values to determine the optimal cessation point for treatment. Severe cases may progress to liver failure, necessitating a liver transplant for survival. Patients who avoid such complications typically experience complete histopathologic resolution of liver damage within approximately 3 months.

Patient education is crucial and should emphasize the importance of storing acetaminophen out of reach of children. Caregivers should be well-informed about appropriate pediatric dosing and the availability of acetaminophen in various formulations for infants, children, and adults. All caregivers should receive guidance on reading medication labels accurately. Furthermore, patients must be educated about the heightened risk of toxicity when combining medications and the importance of avoiding such practices.[38]

## Enhancing Healthcare Team Outcomes

Patients with hepatic failure due to acetaminophen poisoning may require liver transplantation. Early identification and proper management of acetaminophen toxicity are crucial to minimize morbidity and mortality. A comprehensive, multidisciplinary approach is essential to provide patient-centered care and reduce the need for liver transplantation. Healthcare professionals across various specialties, including emergency medicine, critical care, psychiatry, hepatology, toxicology, and surgery, must be familiar with the diagnosis and management of acetaminophen toxicity. This expertise includes recognizing expected clinical presentations, understanding treatment protocols, and ensuring the timely initiation of treatment.

Nurses and pharmacists play vital roles in educating patients and caregivers about the potential toxicity of acetaminophen. A strategic approach based on evidence-based outcomes is necessary to optimize patient results. Effective interprofessional communication is crucial for seamless patient care. By incorporating skill, strategy, and communication principles, healthcare